WORKED SOLUTION

STEP I 2005, Question 11

Mechanics • Time-distance

STEP I 2005, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q11

The key idea

Everything in this question is about how two vectors, $\mathbf{r}$ and $\mathbf{v} = \dot{\mathbf{r}}$, sit relative to each other. Perpendicular means the scalar product $\mathbf{r} \cdot \mathbf{v}$ is zero; parallel means the "cross" combination $xv_y - yv_x$ is zero. Both expressions collapse nicely once we write $\sin 2t = 2\sin t \cos t$ and $\cos 2t = 1 - 2\sin^2 t$. For the distance and the sketch, it pays to write everything in terms of $\cos t$, or to eliminate t altogether.

Solution

Write $\mathbf{r} = x\,\mathbf{i} + y\,\mathbf{j}$ with

$$x = \sin 2t = 2\sin t \cos t, \qquad y = 2\cos t, \qquad 0 \leq t < 2\pi.$$

Differentiating, the velocity is

$$\mathbf{v} = \dot{\mathbf{r}} = (2\cos 2t)\,\mathbf{i} - (2\sin t)\,\mathbf{j}.$$

Part (i)

The particle is at the origin when $x = 0$ and $y = 0$. The condition $y = 0$ means $\cos t = 0$, so $t = \frac{\pi}{2}$ or $t = \frac{3\pi}{2}$ in the given range. At both of these times $x = 2\sin t \cos t = 0$ automatically. So the particle is at the origin at exactly these two times, $t = \frac{\pi}{2}$ and $t = \frac{3\pi}{2}$, and at no others, which is what we had to show.

Part (ii)

The velocity is perpendicular to the displacement when $\mathbf{r} \cdot \mathbf{v} = 0$ (with both vectors non-zero, so that "perpendicular" has a meaning). Now

$$\begin{aligned}\mathbf{r} \cdot \mathbf{v} &= 2\sin 2t \cos 2t - 4 \sin t \cos t \\ &= 2\sin 2t \cos 2t - 2 \sin 2t \\ &= 2 \sin 2t \,(\cos 2t - 1) \\ &= -4 \sin 2t \, \sin^2 t,\end{aligned}$$

using $\cos 2t - 1 = -2\sin^2 t$. This vanishes when $\sin 2t = 0$ or $\sin t = 0$, that is, at

$$t = 0, \ \tfrac{\pi}{2}, \ \pi, \ \tfrac{3\pi}{2}.$$

Ask yourself Should $t = \frac{\pi}{2}$ and $t = \frac{3\pi}{2}$ really be on the list?

No. By part (i), at those times $\mathbf{r} = \mathbf{0}$, and the zero vector has no direction, so it can't be perpendicular to anything in a meaningful sense; the scalar product is zero for a trivial reason. At $t = 0$ we have $\mathbf{r} = 2\mathbf{j}$ and $\mathbf{v} = 2\mathbf{i}$, and at $t = \pi$ we have $\mathbf{r} = -2\mathbf{j}$ and $\mathbf{v} = 2\mathbf{i}$: genuinely perpendicular, non-zero vectors. So the velocity is perpendicular to the displacement at

$$t = 0 \quad \text{and} \quad t = \pi.$$

Part (iii)

First, $\mathbf{v}$ is never zero: $\mathbf{v} = \mathbf{0}$ would need $\sin t = 0$, but then $\cos 2t = 1 - 2\sin^2 t = 1 \neq 0$.

Two non-zero vectors (x, y) and (v_x, v_y) are parallel exactly when $xv_y - yv_x = 0$. Here

$$\begin{aligned}xv_y - yv_x &= \sin 2t \,(-2 \sin t) - 2\cos t \,(2 \cos 2t) \\ &= -4 \sin^2 t \cos t - 4 \cos t \,(1 - 2\sin^2 t) \\ &= -4 \cos t \,(1 - \sin^2 t) \\ &= -4 \cos^3 t.\end{aligned}$$

This is zero only when $\cos t = 0$, which by part (i) happens only when the particle is at the origin. So whenever the particle is away from the origin, $\cos t \neq 0$, the combination is non-zero, and the velocity is not parallel to the displacement. That completes the proof.

Part (iv)

Let $c = \cos^2 t$, so $0 \leq c \leq 1$. Then

$$|\mathbf{r}|^2 = \sin^2 2t + 4\cos^2 t = 4\sin^2 t \cos^2 t + 4\cos^2 t = 4c(1 - c) + 4c = 4c(2 - c).$$

The quadratic $4c(2 - c)$ has its vertex at $c = 1$ and is increasing on $[0, 1]$, so its largest value on this interval is at $c = 1$, giving $|\mathbf{r}|^2 = 4$. The maximum distance from the

origin is therefore

$$|\mathbf{r}|_{\max} = 2,$$

reached when $\cos^2 t = 1$, that is, at $t = 0$ and $t = \pi$, at the points $(0, \pm 2)$. It's reassuring that these are exactly the times found in part (ii): at a maximum of distance, the radial component of the velocity must vanish.

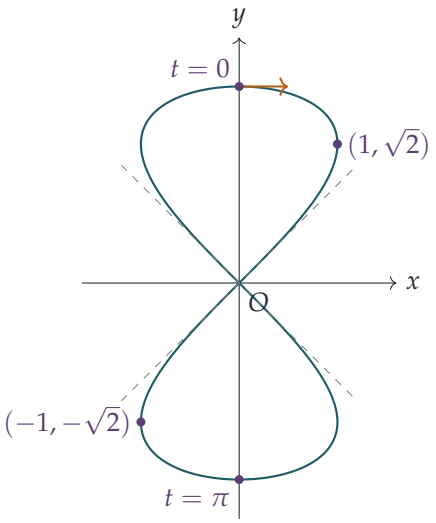
For the sketch, eliminate t. Since $x = y \sin t$ and $\sin^2 t = 1 - \cos^2 t = 1 - \frac{1}{4}y^2$,

$$x^2 = y^2\left(1 - \frac{y^2}{4}\right), \qquad \text{that is,} \qquad 4x^2 = y^2(4 - y^2).$$

Some features to put on the sketch:

- The curve is symmetric in both axes, and $-2 \le y \le 2$.
- $|x| = |\sin 2t| \le 1$, with $|x| = 1$ when $y = \pm\sqrt{2}$.
- Near the origin $4x^2 \approx 4y^2$, so the curve crosses itself there along the lines $y = \pm x$. This matches the velocities $-2\mathbf{i} - 2\mathbf{j}$ at $t = \frac{\pi}{2}$ and $-2\mathbf{i} + 2\mathbf{j}$ at $t = \frac{3\pi}{2}$.
- At $(0, \pm 2)$ the path is horizontal, as part (ii) showed.

Following t: the particle starts at $(0, 2)$ moving in the $+\mathbf{i}$ direction, goes down the right side of the upper loop to the origin, round the lower loop (left side first) back to the origin, and finally up the left side of the upper loop. The path is a figure of eight.



Checks

- At $t = \frac{\pi}{4}$: $\mathbf{r} = \mathbf{i} + \sqrt{2}\,\mathbf{j}$, which satisfies $4x^2 = 4 = y^2(4 - y^2) = 2 \cdot 2$, and $|\mathbf{r}| = \sqrt{3} < 2$, agreeing with $4c(2 - c) = 4 \cdot \frac{1}{2} \cdot \frac{3}{2} = 3$.
- Sampling t finely over $[0, 2\pi)$ gives a largest distance of 2.000, near-zero distances only at $t \approx 1.57$ and 4.71, and confirms the identities $\mathbf{r} \cdot \mathbf{v} = -4 \sin 2t \sin^2 t$ and

$xv_y - yv_x = -4\cos^3 t$ at random times. The smallest speed is about 1.32, so $\mathbf{v}$ is never zero.

Common slips

- Listing $t = \frac{\pi}{2}$ and $\frac{3\pi}{2}$ in part (ii): the scalar product vanishes there only because $\mathbf{r} = \mathbf{0}$.
- In part (iii), forgetting to check that $\mathbf{v}$ is never zero; a zero velocity would be "parallel" to everything.
- Drawing the loops side by side: the eight stands upright, with loops above and below the origin, since y ranges over $[-2, 2]$ but x only over $[-1, 1]$.

Beyond STEP

The quantities $\mathbf{r} \cdot \mathbf{v} = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}|\mathbf{r}|^2$ and $xv_y - yv_x$ are the radial part of the motion and (per unit mass) the angular momentum about the origin; in polar coordinates the second is $r^2\dot{\theta}$. Part (iii) says the angular momentum vanishes only at the origin, so the particle never moves straight towards or away from O. The curve itself is the lemniscate of Gerono, one of many classical figure-of-eight curves. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops polar coordinates and angular momentum in exactly this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.